COVID-19 Global Data Analytics Project

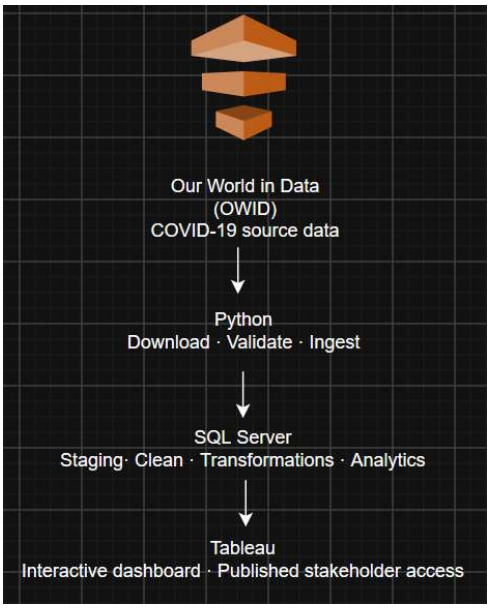
Bradley Fite

CS 621

The COVID-19 Global Data Analytics project will provide public health researchers and healthcare organizations with an accessible, interactive platform for analyzing the global impact of COVID-19. The project will develop an end-to-end data analytics pipeline that retrieves publicly available COVID-19 datasets, processes that information through Python and SQL Server, and presents those analytical findings through an interactive Tableau dashboard. The primary objective is to allow stakeholders the ability to identify trends in reported COVID-19 cases, deaths, vaccination progress, and geographic differences over time.

The intended technology stack includes:

- **Python:** Automated data retrieval, initial validation, and pipeline orchestration
- **SQL Server:** Data storage, cleaning, transformations, and analytical computations
- **Tableau:** Interactive visualization, dashboard publication, and stakeholder-facing delivery



The project will use publicly available COVID-19 datasets from Our World in Data (OWID). This data contains country-level COVID-19 statistics, including reporting cases, deaths, population information, and vaccination metrics.

The relevant fields include:

- *location* – Country or geographic region
- *date* – Date of observation
- *total_cases* – Cumulative reported cases
- *new_cases* – Newly reported cases
- *total_deaths* – Cumulative reported deaths
- *population* – Population of the location
- *total_vaccinations*– Total vaccinations administered

These fields will provide the foundation for further calculations, including population infection percentages, case fatality ratios, and vaccination coverage.

	iso_code	continent	location	date	population	total_cases	new_cases	total_deaths	new_deaths	new_vaccinations	total_vaccinations
728	USA	North America	United States	2022-01-01	338289856	51878860	0	812587	0	120312	521918004
729	USA	North America	United States	2022-01-02	338289856	54590898	2712038	821733	9146	500641	522418645
730	USA	North America	United States	2022-01-03	338289856	54590898	0	821733	0	1252631	523671276
731	USA	North America	United States	2022-01-04	338289856	54590898	0	821733	0	1423448	525094724
732	USA	North America	United States	2022-01-05	338289856	54590898	0	821733	0	1461822	526556546

Starting off, the ingestion stage will use Python to automatically retrieve the COVID-19 datasets from their publicly available source. The Python application will download the source data, validate the files, inspect required columns, and load the records into SQL Server staging tables. Python libraries, pandas and pyodbc, will be integrated to support the file processing and database connection. Validation checks will confirm that the downloaded files contain the

expected fields, the data is readable, and the that the ingestion process is successfully completed. The ingestion pipeline will also include error handling and logging so failures can be identified and corrected. This process will reduce manual file handling and support readable data updates. The estimated work time for this portion of the project is eight hours.

```
Downloading historical COVID-19 dataset...
Download complete: C:\Users\[redacted]CovidDataExplorationProject\data\owid-covid-data.csv
File size: 98,391,483 bytes
SHA-256: 8473d0f0fdf962e1ffbd5b85b18726fc96a49bab109e271186c339725a12b10c
```

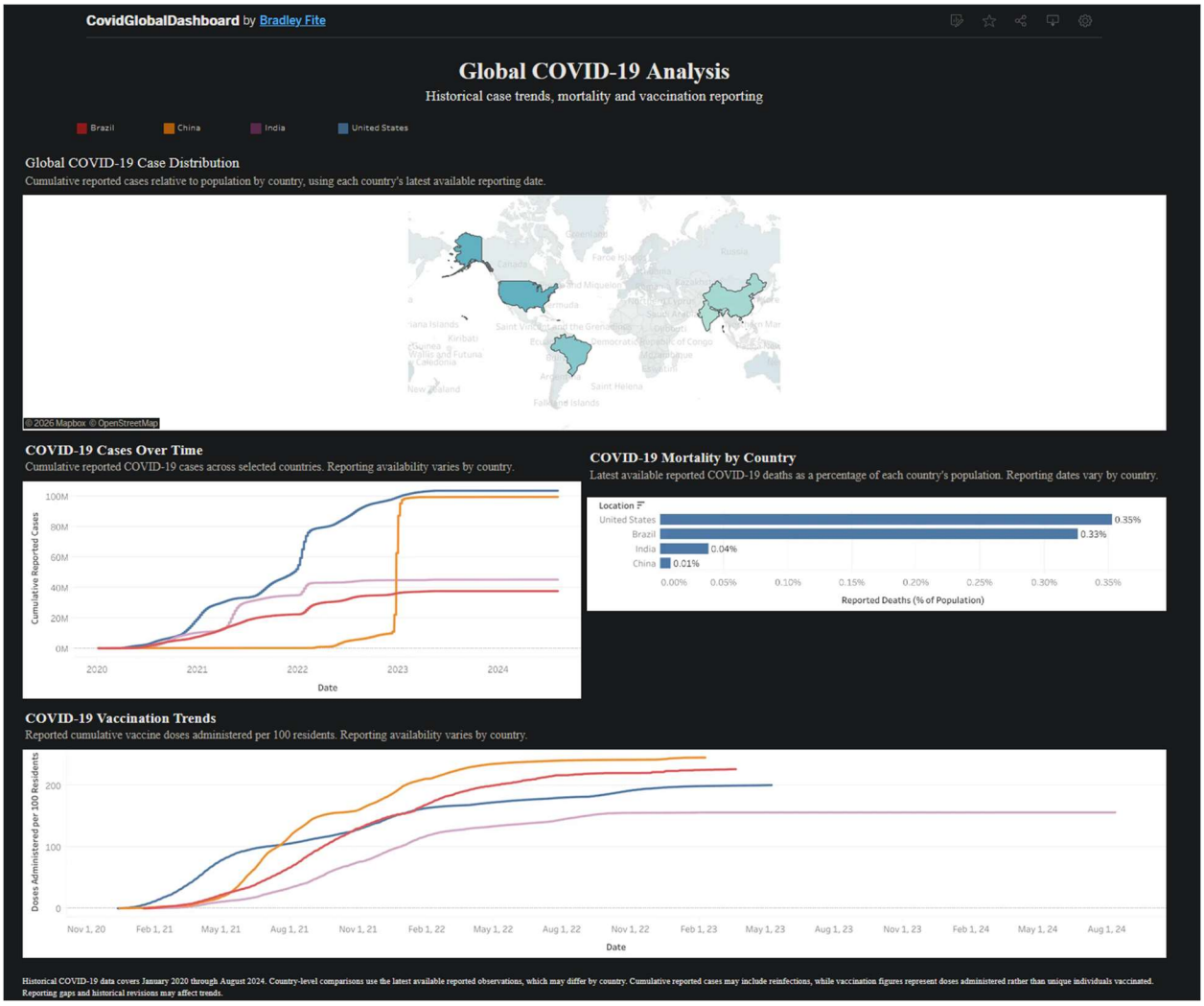
After data ingestion, the COVID-19 data will undergo cleaning and transformation within SQL Server. This part of the project will convert raw source data into reliable, structured datasets suitable for analytical computation and Tableau reporting. This process will use SQL to address missing values, standardize column data types, identify duplicate records, and distinguish individual countries from aggregate geographic categories. Staging tables will preserve the source data while cleaned analytical tables will provide a consistent foundation for downstream calculations. The resulting tables will establish a reusable analytical data model to reduce the amount of complex data preparation required within Tableau itself. The estimated work time for this portion of the project is 16 hours.

iso_code	location	date	continent	population	total_cases	new_cases	total_deaths	new_deaths	new_vaccinations	total_vaccinations
USA	United States	2024-08-04	North America	338289856	103436829	NULL	1193165	619	NULL	NULL
VIR	United States Virgin Islands	2024-08-04	North America	99479	25389	0	132	0	NULL	NULL
USA	United States	2024-08-03	North America	338289856	103436829	NULL	1192546	0	NULL	NULL
VIR	United States Virgin Islands	2024-08-03	North America	99479	25389	0	132	0	NULL	NULL
USA	United States	2024-08-02	North America	338289856	103436829	NULL	1192546	0	NULL	NULL
VIR	United States Virgin Islands	2024-08-02	North America	99479	25389	0	132	0	NULL	NULL
USA	United States	2024-08-01	North America	338289856	103436829	NULL	1192546	0	NULL	NULL
VIR	United States Virgin Islands	2024-08-01	North America	99479	25389	0	132	0	NULL	NULL
USA	United States	2024-07-31	North America	338289856	103436829	NULL	1192546	0	NULL	NULL
VIR	United States Virgin Islands	2024-07-31	North America	99479	25389	0	132	0	NULL	NULL

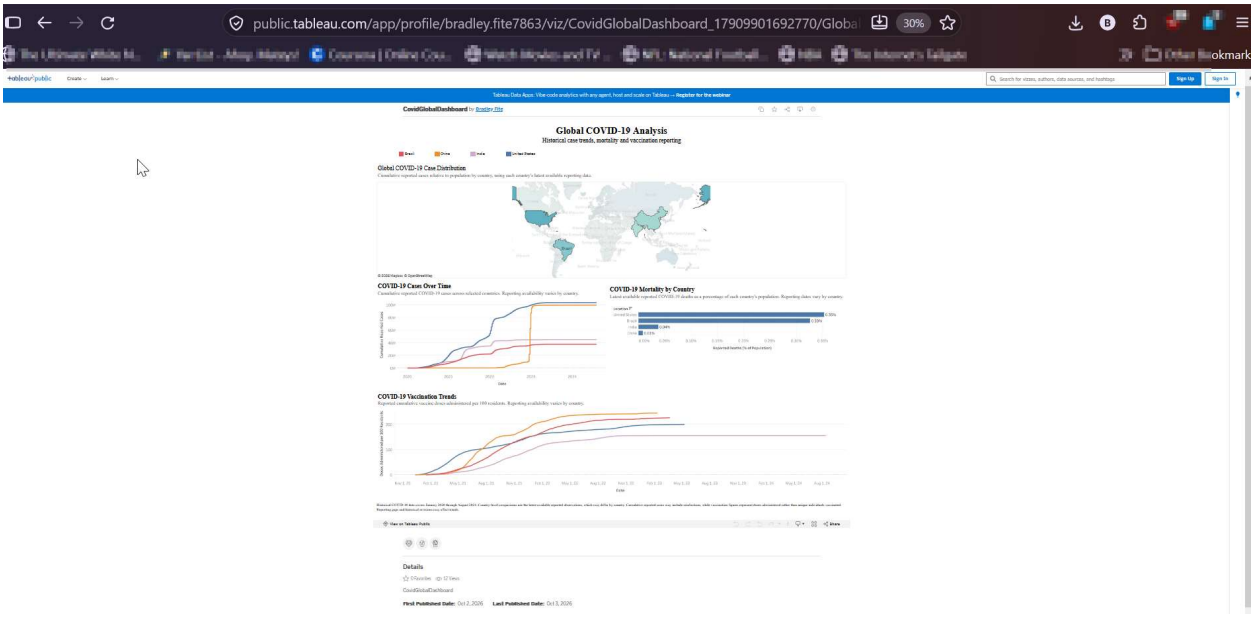
Next, the analytics computation stage will transform the clean COVID-19 data into meaningful metrics that public health researchers and other stakeholders can use to evaluate pandemic trends. SQL queries will calculate cumulative reported infections, cumulative deaths, population infection percentages, case fatality rations, vaccination coverage, and changes in reported cases over time. The analytical results will be stored in reusable SQL Server views specifically designed for ingestion by Tableau. This will centralize metric definitions and ensure that dashboards use consistent calculations in a performance optimized approach. The estimated work time of this portion of the project is 10 hours.

iso_code	continent	location	date	population	new_cases	total_cases	new_deaths	total_deaths	latest_deaths_date	new_vaccinations	total_vaccinations	reported_cases_pct_population	reported_case_fatality_pct	reported_deaths_pct_population
GRC	Europe	Greece	2021-01-28	10384972	0	155772	0	5658	2024-08-04	18245	234285	1.5000	3.6322	0.0545
GRC	Europe	Greece	2021-01-29	10384972	0	155772	0	5658	2024-08-04	19078	253363	1.5000	3.6322	0.0545
GRC	Europe	Greece	2021-01-30	10384972	0	155772	0	5658	2024-08-04	17924	271287	1.5000	3.6322	0.0545
GRC	Europe	Greece	2021-01-31	10384972	5467	161239	167	5825	2024-08-04	NULL	NULL	1.5526	3.6126	0.0561
GRC	Europe	Greece	2021-02-01	10384972	0	161239	0	5825	2024-08-04	NULL	293817	1.5526	3.6126	0.0561

Afterwards, the visualization stage begins, this is where Tableau will be used to transform the SQL Server analytical results into interactive dashboards. Tableau will connect to the validated views, allowing customers to explore COVID-19 trends without directly accessing the underlying database or needing to write SQL queries. The dashboard will contain key performance indicators such as total reported cases, total reported deaths, population infection percentages, and vaccination coverage. Line charts will illustrate changes in reported cases and deaths over time, while geographic maps and comparative charts will highlight differences between countries and region. The dashboard will be designed for public health researchers and healthcare decision-makers, prioritizing clear metric definitions, readable visuals, and straightforward navigation. The estimated work time for this portion of the project is 14 hours.



The primary platform for delivery of this project to the stakeholders and public health researchers will be Tableau. The dashboard will be interactive and give healthcare organizations access to analyzed pandemic data. It will be published on Tableau Cloud or Tableau Server, allowing authorized customers to access the visualization through a web browser without needing to install extra software. The data pipeline will support periodic updates from the source datasets, enabling a data source refresh cadence that meets the needs of those viewing and using the dashboard. Customers will be able to filter data, interact with visualizations, and export supported views or summarized results for further research.


public.tableau.com/app/profile/bradley.fite7863/viz/CovidGlobalDashboard_17909901692770/Globa
30%
Global COVID-19 Analysis
Historical case trends, mortality and vaccination reporting
Global COVID-19 Case Distribution
COVID-19 Cases Over Time
COVID-19 Mortality by Country
COVID-19 Vaccination Trends
Details
CovidGlobalDashboard
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